



Association between outdoor jogging behavior and PM_{2.5} exposure: Evidence from massive GPS trajectory data in Beijing

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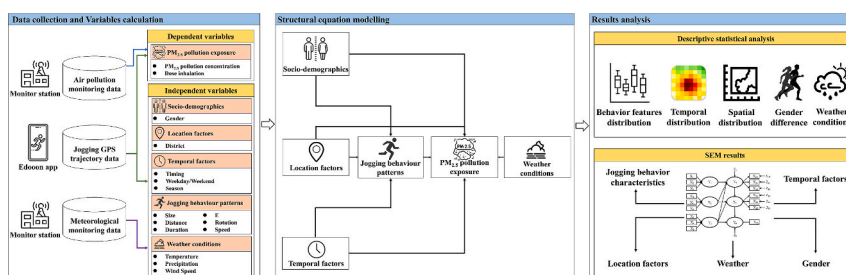
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HIGHLIGHTS

- Joggers' PM_{2.5} exposure was assessed in both exposure concentration and dose inhalation.
- Joggers' PM_{2.5} exposure varies across locations, timings of the day, days of the week and seasons.
- Female joggers inhaled significantly less PM_{2.5} dose than men.
- Jogging behavior patterns mediate the effects of temporal and spatial factors on PM_{2.5} exposure.
- The findings provide values for behavioral interventions in promoting jogging.

GRAPHICAL ABSTRACT



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ABSTRACT

Outdoor jogging is one of the most popular practised exercises worldwide, providing various benefits for health and wellbeing. However, PM_{2.5} exposure risks of jogging behaviors were rarely explored. This study aims to investigate the association between jogging behavior and PM_{2.5} exposure with big data. PM_{2.5} exposure concentration and dose inhalation of individuals were calculated by integrating hourly PM_{2.5} concentration data and jogging GPS trajectory recorded by a sports app during 2015 in Beijing, after which relationships between jogging behaviors and PM_{2.5} exposure were unpacked using statistics analysis and structural equation modelling. Experimental results on massive jogging trajectories show that: (1) the average jogging PM_{2.5} exposure concentration is 60.43 µg/m³, and female joggers inhaled significantly less air pollution dose (19.70 µg) than men (24.91 µg). (2) There exist significant spatiotemporal disparities in jogging exposure to PM_{2.5}. Joggers in the city center, in the morning, on weekdays and in autumn and winter seasons were exposed to higher pollution concentrations. (3) Jogging behavior characteristics, especially distance, activity space size, duration and rotation, were systematically associated with PM_{2.5} exposure across space and time. (4) The role of gender directly shaped joggers' dose inhalation of PM_{2.5} pollution and indirectly via duration, timing choice and distance. (5) The effects of weather conditions on joggers' exposure to PM_{2.5} are mainly via direct effects, whereas the direct impacts of precipitation and wind speed are mitigated by indirect effects stemming from jogging behavior patterns. Our findings provide insights for personal guidance and policy intervention for the sake of promoting physical activity and reducing PM_{2.5} exposure.

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1. Introduction

The pandemic of physical inactivity has become the fourth global health problem (Althoff et al., 2017; Remme et al., 2021). Insufficient physical activity causes 830,000 deaths annually and has been associated with several non-communicable diseases (Dong et al., 2023; Yang et al., 2024a), including cardiovascular disease, stroke, cancer and diabetes. Policy actions have been proposed to promote walking, jogging, cycling and other recreational activities as well as investment in emissions reduction and cleaner air (Remme et al., 2021; World Health Organization, 2021). Outdoor fitness jogging is a simple and highly scalable form of physical activity as it does not require special skills or equipment (Dong et al., 2023; Yang et al., 2024a), which has become a popular ‘lifestyle sport’ among urban residents (Yang et al., 2023; Huang et al., 2023a). With the popularity of outdoor jogging that led to a great variety of geographical contexts (Huang et al., 2023a; Liu et al., 2022; Liu et al., 2023), e.g., along roads, parks and sports squares, jogging individuals are inevitably exposed to certain levels of environmental risks (air pollution, noise, etc.) (Kan et al., 2023; Marquart et al., 2022).

Ambient (outdoor) air pollution is one of the greatest environmental risks to human health (Poom et al., 2021; Hou et al., 2022) and is estimated to cause 4.2 million premature deaths worldwide per year (World Health Organization, 2021). This mortality is due to exposure to fine particulate matter (airborne particles with aerodynamic diameter equal to or $<2.5\ \mu\text{m}$), which can penetrate deep into the lungs and bloodstreams unfiltered and cause heart disease, lung cancer, and both chronic and acute respiratory diseases (World Health Organization, 2021; Reggente et al., 2015). The updated WHO Air Quality Guidelines (World Health Organization, 2021) state annual average concentrations of $\text{PM}_{2.5}$ should not exceed $5\ \mu\text{g}/\text{m}^3$, while 24-h average exposures should not exceed $15\ \mu\text{g}/\text{m}^3$, and there is no safe value of $\text{PM}_{2.5}$ concentration for human health (World Health Organization, 2021).

Scholars from wide disciplines have assessed human exposure to air pollution (Zhong et al., 2023; Mueller et al., 2022; Al-sareji et al., 2022; Xu et al., 2021). Spatial epidemiologists have mainly focused on the residential/location-based assessment of air pollution exposure, where people’s spatiotemporal movement was ignored (Reggente et al., 2015). Numerous mobility-based studies (Al-sareji et al., 2022; Xu et al., 2021; Bigazzi and Figliozzi, 2014; Martínez et al., 2023) have analysed air pollution concentrations in different transport environments using field experiments. Prior works (Bigazzi and Figliozzi, 2014; Martínez et al., 2023) found that walking and cycling environments were generally of lower levels of $\text{PM}_{2.5}$ air pollution concentration than those of car and bus users. However, these results are not always consistent across geographical locations, as Northover (Northover, 2017) found that both cyclists and pedestrians were more exposed than car occupants for $\text{PM}_{2.5}$ (car-to-walk ratio 0.7, car-to-bike ratio 0.9). In addition, the inhalation rates during physical activities such as outdoor fitness jogging were not considered, where the inhalation rates during moderately intensive physical activities could be two to three folds of sitting or standing (Bigazzi and Figliozzi, 2014). With the implication of big data (Liu et al., 2023; Xu et al., 2021), mobility-based exposure to air pollution could be assessed on a large scale in exploring its relationship with people’s movement patterns. Moreover, location-based assessments could lead to more errors than mobility-based approaches when measuring people’s actual exposure to air pollution (Song and Kwan, 2023), which requires new data and methods to measure air pollution exposure of outdoor jogging, such as GPS trajectory data and crowdsourced geographic information (Huang et al., 2023a; Liu et al., 2022; Marquet et al., 2022; Sun et al., 2017).

In light of these, this study aims to assess jogging-related $\text{PM}_{2.5}$ exposure, both in terms of exposure concentration and dose inhalation, and to systematically examine the associations between jogging behavior and $\text{PM}_{2.5}$ exposure across space and time. Massive jogging GPS trajectory data and hourly air pollution concentration data in Beijing from Jan 1, 2015, to Dec 31, 2015, were utilized to assess jogging-

related $\text{PM}_{2.5}$ exposure at the individual level. Then, the spatiotemporal disparities in jogging behavior patterns and related $\text{PM}_{2.5}$ exposure were investigated using descriptive statistics. Lastly, structural equation modelling (SEM) was applied to unpack the systematic relationship between jogging behavior patterns and $\text{PM}_{2.5}$ exposure in different locations and across different temporal frames.

The remainder of this paper is organized into five sections. Section 2 reviews relevant research articles and highlights the research gaps to be addressed. Section 3 introduces the study area, datasets and methods. Section 4 provides the experimental results analysis. Section 5 and Section 6 discuss the main findings, limitations and future directions.

2. Related work

2.1. Assessing air pollution exposure for behavior studies

Exposure assessment serves as a vital instrument for quantitatively elucidating the extent of individuals’ actual exposure to air pollutants (Song and Kwan, 2023; Song et al., 2019). Numerous research (Reggente et al., 2015; Zhong et al., 2023; Mueller et al., 2022; Al-sareji et al., 2022; Xu et al., 2021; Bigazzi and Figliozzi, 2014; Martínez et al., 2023) have been conducted on this topic, which can be roughly categorized into two types: assessment based on point-fixed activities (Jeong et al., 2019; Panunzi et al., 2023; Nishida and Yatera, 2022; Yoo et al., 2023) and assessment based on individuals’ trajectories (Ma et al., 2021; Park, 2020; Kwan et al., 2014; Yu et al., 2020). Studies based on point-fixed activities mainly assigned individual activities to a point location (e.g., residential address, workspace) (Jeong et al., 2019; Panunzi et al., 2023; Nishida and Yatera, 2022; Yoo et al., 2023), and then examined the exposure level with uniform or variable air quality. Air pollution concentration at specific locations was often estimated based on air pollution modelling (statistical interpolation, land use regression models, etc.) for both population-level and individual-level assessment (Song et al., 2019) or direct measures using air pollution monitors for individual-level assessment (Jeong et al., 2019). The associations of population/individuals’ exposure to air pollution at fixed locations with physical health (chronic and acute respiratory diseases (Panunzi et al., 2023), occupation diseases (Nishida and Yatera, 2022), etc.) and psychological health were further discussed. In addition, there is growing research into the social disparities in point-fixed air pollution exposure among various racial and ethnic groups (Yoo et al., 2023). However, these studies based on static locations ignored the dynamics of people’s movement, which could lead to an imprecise assessment of air pollution and induced health outcomes.

Another group of individual-level assessments consider people’s movement trajectory and spatiotemporally varied air pollution (Ma et al., 2021; Park, 2020). Individuals’ movement was usually derived from travel logs or trajectory data, after which air pollution exposure was measured by combining air pollution concentration data from monitor stations or portable air pollution sensors (Ma et al., 2021). Following the dynamic air pollution assessment framework (Kwan et al., 2014), researchers have incorporated (GPS-enabled) travel diaries and hourly air pollution concentration from monitor stations in assessing individuals’ exposure to air pollution (Park, 2020). Ma et al. (Ma et al., 2021) assessed individuals’ real-time air pollution exposure using a travel diary that requires participants to carry a portable air pollution sensor along their travels and activities. However, these studies are often limited to small sample sizes due to high time and cost demands for questionnaire-based surveys.

Geospatial big data approaches could provide massive trajectory data to estimate individuals’ exposure to air pollution on a large scale (Huang et al., 2023a; Song et al., 2019). Nyhan et al. (Nyhan et al., 2019) quantified people’s air pollution exposure in the Greater Boston Area using mobile phone data in combination with $\text{PM}_{2.5}$ data from 71 monitor stations. However, the trajectories inferred from cellular network data could be erroneous in representing people’s actual

movement (Yu et al., 2020). Massive GPS tracking data regarding walking and cycling (Mueller et al., 2022; Xu et al., 2021) are used to assess individuals' PM_{2.5} exposure for specific types of physical activity. It is still difficult to collect GPS tracks of jogging behavior on a large scale, resulting in a lack of work on assessing outdoor jogging exposure to air pollution.

Despite the substantial body of research that has been generated, the primary focus has been on assessment based on point-fixed locations or travel diaries, and assessment for active travel (e.g., walking and cycling). To the best of our knowledge, there is still no work that addresses outdoor jogging for leisure and fitness purposes, especially fine-grained, and real-time dynamic measuring individual outdoor jogging-related PM_{2.5} exposure in large urban areas. Therefore, there is a need to integrate crowdsourced geographic information, such as GPS trajectory data and real-time air quality monitoring data, for the dynamic assessment of PM_{2.5} exposure and inhalation levels during outdoor jogging.

2.2. Association between physical activity and air pollution exposure

Human activities and environmental exposures (air pollution, noise, green spaces, built environments, etc.) have become the focal point of multidisciplinary research in fields such as geography (Kan et al., 2023; Marquart et al., 2022), behavioral science (Song and Kwan, 2023; Sun et al., 2017), transportation (Zhong et al., 2023; Xu et al., 2021) and public health (Panunzi et al., 2023; Nishida and Yatera, 2022). A wealth of research on human activity and air pollution exposure continues to emerge, indicating significant associations of air pollution exposure with activity types (Ma et al., 2021; Park, 2020), locations (Nishida and Yatera, 2022), transport modes (Al-sareji et al., 2022), trip purpose (Park, 2020), etc. Most studies investigated the association between transport modes and air pollution have focused on motorized transport (Zhong et al., 2023), with a few studies looking at walking and cycling, as active travel modes for physical activities (Mueller et al., 2022) and even fewer considering other forms of physical activities.

Previous works mainly focused on physical activities, e.g., walking and cycling (Bigazzi and Figliozzi, 2014; Sun et al., 2017) in this topic. For example, PM_{2.5} exposure during walking and cycling was found not consistently lower compared with motorized transport microenvironments across studies in the global North (Marquart et al., 2022; Bigazzi and Figliozzi, 2014). Huang et al. (Huang et al., 2012) reported cyclists were exposed to higher levels of PM_{2.5} concentration (49.1 µg/m³) compared with bus riders (42.4 µg/m³) and car occupants (31.6 µg/m³). The exposure for cyclists and pedestrians could be even higher when inhalation rates and trip duration were taken into consideration (Huang et al., 2012). Here, the dose of inhaled air pollution is critical to examine the dose-response relationship between air pollution exposure and induced health problems, especially while doing physical activities (Tainio et al., 2021). The pollutant inhalation rate is the product of exposure concentration and inhalation rate, measured in the volume of air per unit time (usually L/min) (Tainio et al., 2021). Moreover, the route choice of cyclists and pedestrians could significantly affect their exposure to air pollution (Bigazzi and Figliozzi, 2014; Li et al., 2017). For example, prior work (Bigazzi and Figliozzi, 2014) found that the median PM_{2.5} exposure concentration in high-traffic environments was 26 % higher than in low-traffic environments. Li et al. (Li et al., 2017) found a 3.2 % reduction in inhaled dose for PM_{2.5} on the cleanest versus shortest pedestrian path.

Despite many efforts, there are relatively few studies on outdoor fitness jogging due to data limitations. In recent years, with the development of crowdsourced geographic information, outdoor jogging behavior patterns (Tian et al., 2022) and their associations with multiple built environment factors have been increasingly explored using the jogging GPS tracking data recorded by sports apps (Yang et al., 2023; Huang et al., 2023a; Liu et al., 2022; Liu et al., 2023), such as Strava (Yang et al., 2023; Liu et al., 2023) and Edoon (Yang et al., 2024a;

Huang et al., 2023a). To the best of our knowledge, the associations between jogging behavior patterns and PM_{2.5} exposure have not been studied.

In summary, there are gaps to be bridged for the studies associating outdoor fitness jogging and PM_{2.5} exposure: (1) Most related studies examined the association between physical activity and PM_{2.5} exposure have focused on walking and cycling. The relationship between outdoor fitness jogging and PM_{2.5} exposure (exposure concentration and dose inhalation) is not explored. (2) Previous studies mainly focused on PM_{2.5} exposure assessment of physical activity, while ignoring the associations between multiple dimensions of behavior features and PM_{2.5} exposure. Our work is the first study to assess outdoor fitness jogging exposure to PM_{2.5} and to systematically examine the associated spatial, temporal, behavioral, and demographic factors in shaping these exposures.

3. Materials and methods

3.1. Study area

Beijing, the capital of the China, is located in the north of the North China Plain ranging from 115°25'E to 117°30'E longitude and from 39°28'N to 41°05'N latitude (Fig. 1). The central city of Beijing, including six main districts (Dongcheng, Xicheng, Haidian, Chaoyang, Fengtai and Shijingshan), with a total area of 1368.32 km² and a population of 12.6 million in 2015 (Tian et al., 2022; Wu and Kim, 2021) was selected. These areas have well-established leisure or sports facilities and green spaces (Wu and Kim, 2021) since experiencing rapid economic growth and urban expansion in recent years. However, the increasingly intensive human activities have brought tremendous impacts to the environment, e.g., air pollution. The annual PM_{2.5} average concentration in Beijing is 80.6 µg/m³, with the highest monthly average value of 152.1 µg/m³ in December and the lowest monthly average value of 44.7 µg/m³ in August (Beijing Municipal Ecology and Environment Bureau, 2024). PM_{2.5} air quality accedes to the annual standard of 15 µg/m³ standard according to WHO air quality guidelines (World Health Organization, 2021).

3.2. Data collection and processing

3.2.1. Jogging GPS trajectory data

This work utilized the GPS trajectory data of joggers recorded by the Edoon Sports App (Yang et al., 2023; Liu et al., 2022; Tian et al., 2022). Edoon App, an outdoor sports fitness application in China, had a large number of users from 2013 to 2017 (Yang et al., 2023; Tian et al., 2022), on par with Gudong and Yuepao Apps. Until March 2016, Edoon's downloads reached 5.79 million, closely rivalling the download numbers of Gudong (6.51 million) and Yuepao (8.32 million), which is one of the most popular apps for fitness and jogging in the outdoor contexts (Liu et al., 2022; Tian et al., 2022). The data contains an anonymous user ID, user gender (incomplete), and high space-time resolution GPS points with longitude, latitude, and time tags. The time resolution ranges from 1 s to 10 s and the average time resolution is 4–5 s. The position error of the GPS point is about 10–50 m. Trajectory data from the Edoon App has been widely used in behavior patterns mining (Tian et al., 2022) and built environment-behavior relationship analysis (Yang et al., 2023; Liu et al., 2022).

Jogging trajectories pre-processing mainly includes the following steps. (1) Processing trajectory noise. Following prior work (Lee and Krumm, 2011), we performed a median filter to smooth noise points (e.g., the distance between two consecutive GPS points was larger than 200 m). (2) Removing abnormal jogging trips. We removed trajectories with abnormal distances (e.g., >30 km, lower than 0.8 km) and abnormal average speeds (e.g., higher than 25 km/h, lower than 3.7 km/h) (Tian et al., 2022). Thus, some abnormal trajectories with speeds lower than normal, including stop trajectories and indoor trajectories,

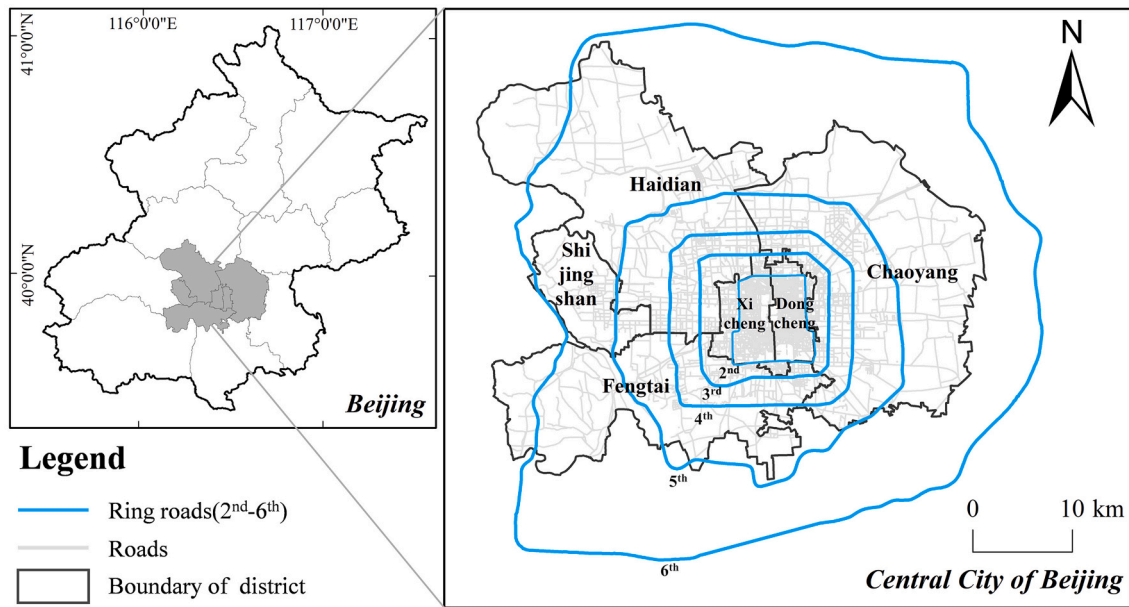


Fig. 1. Study area.

will be filtered out. (3) Completing user gender information. Referring to the existing works (Wu et al., 2019; Liu et al., 2020; Zhang et al., 2020), we utilized a random forest model and jogging GPS trajectory data to infer the missing gender information for some users. In our dataset, 14,858 users have gender data with 9883 males and 4975 females, and 13,860 users have no gender information. Based on the existing works (Wu et al., 2019; Liu et al., 2020; Zhang et al., 2020), we selected 14 jogging trajectory behavior features to train the prediction model. For each user, 14 behavior features were extracted from jogging trajectories, including the number of trips, activity frequency on weekdays and weekends, average speed, average duration, average distance, jogging frequency during the morning (5:00–11:00), midday (11:00–17:00), evening (17:00–22:00), and night periods (other periods), activity space size (calculating based on standard deviation ellipse using jogging trajectory points (Ma et al., 2021)), maximum trip distance, minimum trip distance, and median of jogging trip distance. Taking these extracted features as input, we trained and established the random forest classification model through supervised learning, implemented by the Python library (Scikit-Learn) (Liu et al., 2023; Liu et al., 2020). Specifically, the training sample dataset was split into a training set and a test set in a 7:3 ratio. Then, the Grid Search method and 10-fold cross-validation (Liu et al., 2023) were adopted to obtain the optimal hyper-parameters and to prevent overfitting (Liu et al., 2023). Finally, the accuracy, precision and F1 score of the prediction results by the random forest achieved were 0.702, 0.713 and 0.824, respectively. These results are comparable to the results in the prior works (Wu et al., 2019; Liu et al., 2020; Zhang et al., 2020), demonstrating the effectiveness of the prediction model.

By using the prediction model for gender data completion, there are a total of 22,002 male and 6716 female users in our dataset. In the raw jogging trajectory dataset, there are a total of 301,792 trajectory trips with 103,660,689 GPS points. After data pre-processing, we extracted 287,663 trips (male: 252,763; female 35,900) with 96,939,911 GPS points involving 28,718 users (male: 22,002; female: 6716) from January 2015 to December 2015 for experiment analysis. The raw jogging GPS trajectories within the study are shown in Fig. 2.

3.2.2. Air quality data

PM_{2.5} is selected as the indicator of air pollution, as PM consists of a complex mixture of sulfate, nitrates, ammonia, sodium chloride, black carbon, mineral dust and water and affects more people than any other pollutant, and PM_{2.5} can penetrate the lungs and enter the blood

circulation system, contributing to the risk of developing multiple severe diseases (World Health Organization, 2021). Calculating individual PM_{2.5} exposure requires PM_{2.5} concentration data with high spatial and temporal resolution. Following prior works (Lee et al., 2017; Zhao et al., 2024; Hu et al., 2023), we fuse two PM_{2.5} datasets to produce hourly PM_{2.5} concentration data for each 1-km grid.

One dataset is the hourly PM_{2.5} concentrations from 35 air quality monitoring stations from Jan. 2015 to Dec. 2015, which were collected from the Beijing Municipal Ecological and Environmental Monitoring Center (www.bjmemc.com.cn). The spatial distribution of the monitoring stations is shown in Fig. 3(a). The other open dataset is China High Air Pollutants (CHAP), a high spatial resolution and high-quality near-surface PM_{2.5} dataset for China, reconstructed by Jing et al. (Wei et al., 2020; Wei et al., n.d.; Wei and Li, 2019). They developed a Space-Time Extra-Trees (STET) model to estimate PM_{2.5} concentrations at a 1 km resolution by integrating aerosol optical depth (AOD) data, meteorological data, land surface conditions, and population distribution (Wei et al., 2020; Wei et al., n.d.). The model used to estimate PM_{2.5} in this dataset can capture well variations in PM_{2.5} concentrations, with higher accuracies (i.e., cross-validation coefficient of determination, $CV-R^2 = 0.86\text{--}0.90$), stronger predictive powers (i.e., $R^2 = 0.80\text{--}0.82$) and a lower root mean square error (i.e., $10.33\text{ }\mu\text{g m}^{-3}$) (Wei et al., 2020; Wei et al., n.d.). This dataset offers the daily average PM_{2.5} surface concentrations on a 1-km grid across China from 2000 to 2021 (Wei and Li, 2019). In our work, we use daily PM_{2.5} concentration data in Beijing, spanning Jan. 2015 to Dec. 2015, within 1-km grids (Fig. 3) to generate a new high-resolution PM_{2.5} data.

The PM_{2.5} data from the monitoring stations has high temporal resolution but low spatial resolution, whereas the CHAP dataset has high spatial resolution but low temporal resolution. Therefore, following previous studies (Lee et al., 2017; Zhao et al., 2024; Hu et al., 2023; Wei et al., 2020), we utilize hourly PM_{2.5} data collected by fixed monitoring stations to correct the daily average PM_{2.5} data of 1-km grid from the CHAP dataset, generating a new dataset that hourly PM_{2.5} concentration data for each 1-km grid in Beijing. Specifically, the correction factor is proposed to obtain the hourly PM_{2.5} concentration in each grid according to prior works (Lee et al., 2017; Zhao et al., 2024; Hu et al., 2023; Wei et al., 2020). For a monitoring station s , we use $C_{s,d,h}$ to represent its PM_{2.5} concentration on day d hour h , and the correction factor $CF_{s,d,h}$ of the station at the h hour of the d day is calculated as follows:

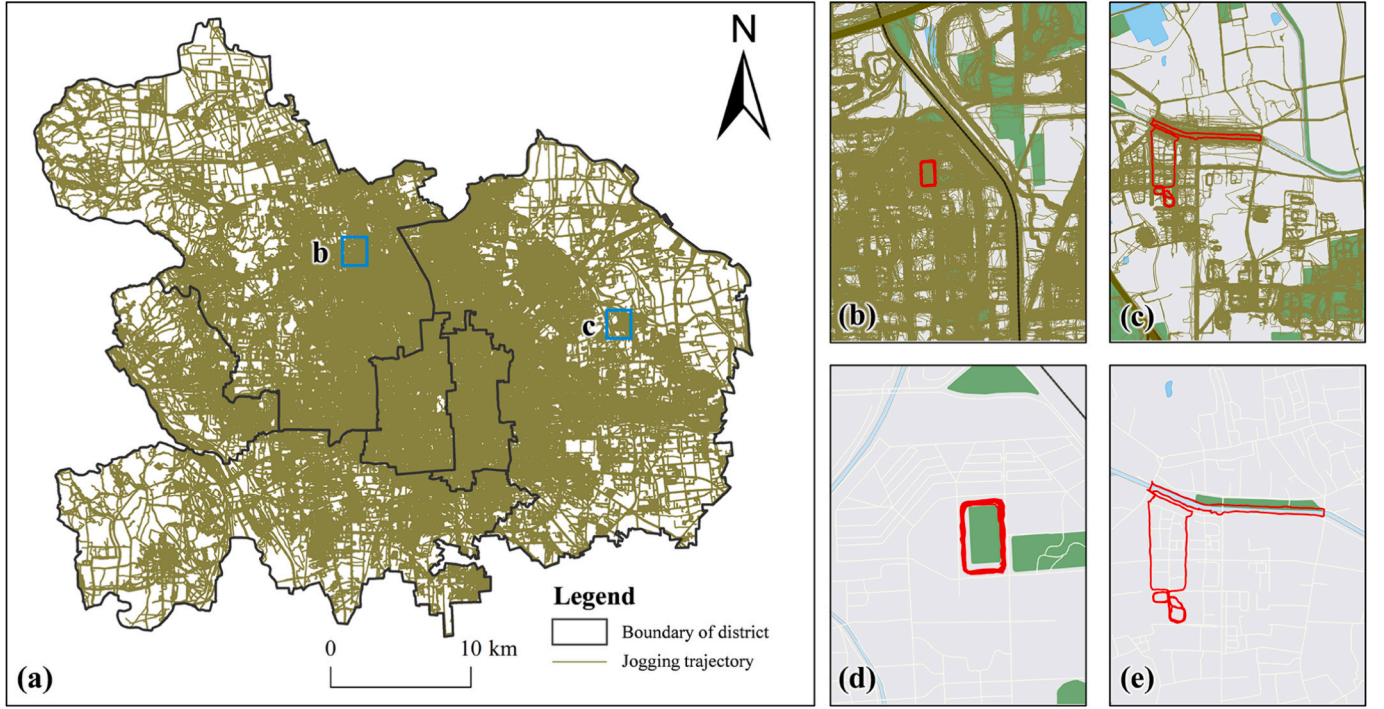


Fig. 2. Jogging GPS trajectory data in (a) the study area, (b) Tsinghua University, (c) Dongba community, (d) a playground, and (e) a street.

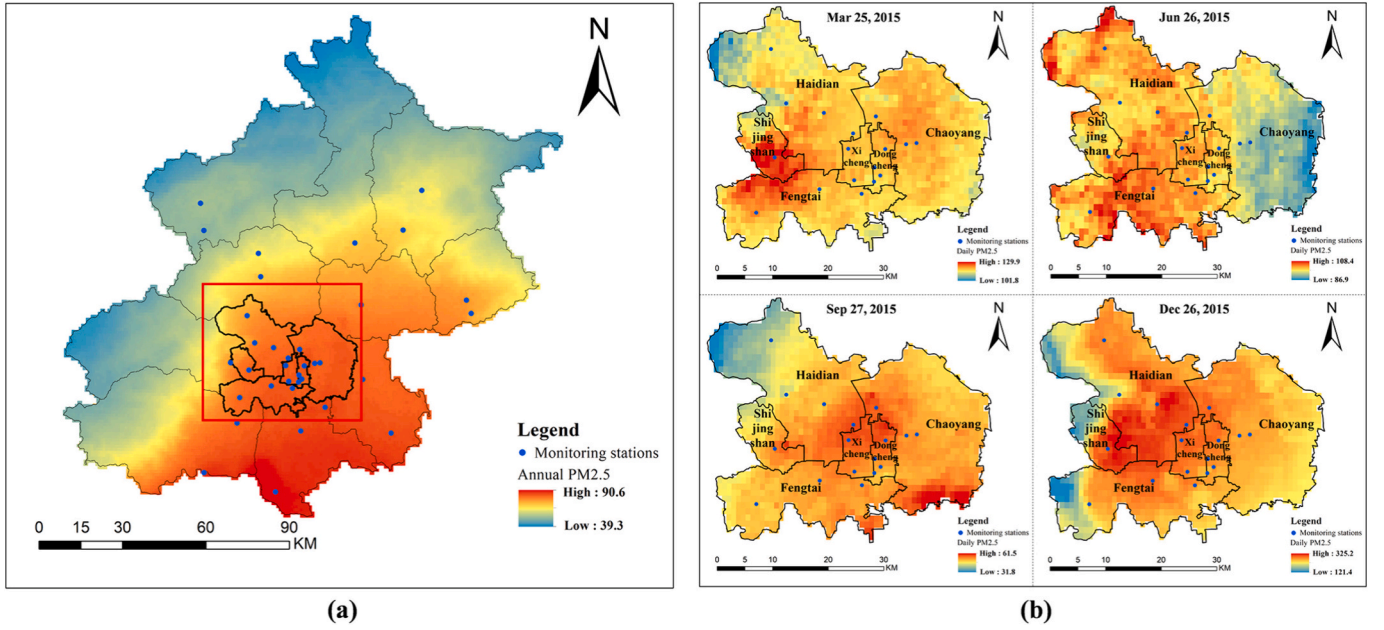


Fig. 3. Two PM_{2.5} datasets; (a) monitoring station locations and (b) daily average PM_{2.5} concentration in Beijing from CHAP dataset.

$$C_{s,d} = \frac{\sum_{h=0}^{23} C_{s,d,h}}{24} \quad (1)$$

$$CF_{s,d,h} = \frac{C_{s,d,h}}{C_{s,d}} \quad (2)$$

where $C_{s,d}$ represents the hourly average PM_{2.5} concentration at station s on day d ; $CF_{s,d,h}$ is the correction coefficient for the nearby grids of station s at hour h on day d . We can calculate a total of $35 * 24 * 365 = 306,600$ correction factors since there are 35 monitoring stations in

Beijing. Then for every grid, we use the correction factor of the monitoring station that has the nearest distance to it to correct its PM_{2.5} concentration data and obtain the hourly average PM_{2.5} concentration in this grid. The formula is defined as follows:

$$\hat{C}_{(x,y),d,h} = CF_{s,d,h} \times \hat{C}_{(x,y),d} \quad (3)$$

where $\hat{C}_{(x,y),d}$ is the daily PM_{2.5} concentration of grid (x,y) in the d -th day from the CHAP dataset; $CF_{s,d,h}$ is the correction factor; $\hat{C}_{(x,y),d,h}$ is the hourly PM_{2.5} concentration of grid (x,y) in the h -th hour on d -th day. Finally, we have obtained the fused high-resolution PM_{2.5} concentration

(per hour for each 1-km spatial grid) of 24 h per day for the entire year of 2015.

The fused PM_{2.5} data are evaluated from both qualitative and quantitative perspectives in Fig. 4. Fig. 4(a) shows significant spatio-temporal continuity features and dynamic variations in air pollution diffusion across different hours. This indicates that the fused hourly PM_{2.5} data simulate real conditions. Moreover, the PM_{2.5} datasets of the 35 monitoring stations are randomly divided into a test set (80 %) and a validation set (20 %) to evaluate the accuracy of fused results. Specifically, we randomly select 28 monitoring stations (totaling 35 stations) for data fusion, and use the ground truth from the remaining 7 stations for accuracy verification. The fused hourly PM_{2.5} concentration (all hours of 2015) is compared with the observed value of the 7 monitor stations in Fig. 4(b). Through this cross-validation process, it can be seen from the figures that the goodness of fit of most hours is above 0.82, which indicates that the fused PM_{2.5} data has high accuracy and effectiveness. In fact, the accuracy of our experimental data is higher than the validation results in Fig. 4(b) because 35 monitoring stations were used for data fusion correction.

3.2.3. Meteorological data

Meteorological data from Jan 1, 2015, to Dec 31, 2015, including temperature, precipitation and wind speed at hourly intervals (Yang et al., 2024b), was derived from the Beijing Municipal Bureau of Meteorology (<http://bj.cma.gov.cn/>). This dataset is used to derive the weather indicators.

3.3. Variables calculation

3.3.1. Dependent variables: PM_{2.5} exposure

We defined air pollution (PM_{2.5}) exposure into two indicators: average air pollution exposure concentration (Air pollution exposure indicator 1, APE₁) and air pollution dose inhalation (Air pollution exposure indicator 2, APE₂). APE₁ provides information about how much PM_{2.5} pollution is present within specific areas along joggers' paths, while APE₂ calculates how long joggers are exposed to certain levels of PM_{2.5} pollution concentration and quantifies the dose of pollutants that individuals inhale into their respiratory systems. Together, these indicators offer valuable insights into the nature and extent of PM_{2.5} exposure and help inform strategies for mitigating its adverse effects on health.

By integrating jogging GPS trajectory data and hourly PM_{2.5} data, the two indicators were calculated as the following steps (Ma et al., 2021): (1) Generating a PM_{2.5} distribution map. PM_{2.5} concentration map was produced via the method in Section 3.2.2, and thus 8760 layers of PM_{2.5} maps were generated. (2) Assigning PM_{2.5} concentration to GPS points. PM_{2.5} concentration for each GPS point was calculated according to its location and time based on the maps of hourly PM_{2.5} pollution. (3) Calculating the two indicators. APE₁ was obtained as the average value of the PM_{2.5} concentration of each GPS point for each trajectory trip, whilst APE₂ was the weighted summation of PM_{2.5} concentration at each GPS point multiplying the duration between each two GPS point. The formulas are defined as follows:

$$APE_1 = \sum_i d_{ij} \cdot (t_j - t_{j-1}) / \sum_i t_j \quad (4)$$

$$APE_2 = \sum_i d_{ij} \cdot (t_j - t_{j-1}) \cdot I \quad (5)$$

where d_{ij} represents the PM_{2.5} density of a jogging track point j at a certain point of time t ; t_j, t_{j-1} denote the time stamp of a jogging track point j and $j-1$. I is the constant air inhalation rate under jogging conditions, i.e., male: 9.4 L/min; female 7.6 L/min under lighted labour activities (World Health Organization, 2021). Therefore, the average exposure concentration and the total inhalation of PM_{2.5} were

considered.

3.3.2. Independent variables

We characterized the behavior patterns of outdoor jogging from movement, spatial (location) and temporal (timing of the day, weekday/weekend and seasonal) dimensions. Referring to prior works (Ma et al., 2021; Park, 2020; Liu et al., 2020; Zhang et al., 2020; Yuan et al., 2012), a series of dependent variables were designed to represent the movement features of outdoor jogging, including activity space size, distance, duration, rotation, eccentricity (E) and speed. Activity space size reflects one's capability to jog around the projected activity area, where the concentration could vary to a greater degree when they reach further places (Marquet et al., 2022). The rotation of activity space represents the overall travel direction of a jogger. It is included as PM_{2.5} concentration often exhibits directional distribution patterns across space and time (Huang et al., 2023b; Su and Goulias, 2023) and there may exist complex interrelations between PM_{2.5} diffusion and jogging behavior. Eccentricity illustrates the shape of activity space, influencing exposure to varying levels of PM_{2.5}. Distance, duration and speed are fundamental characteristics of jogging behavior that indicate the intensity of physical activity in the spatial path, time consumption and inhalation rate (Wu and Kim, 2021). These parameters are closely linked to PM_{2.5} exposure concentration and dose inhalation.

These indicators were calculated from jogging trips that are generated from real-time tracking GPS points (Ma et al., 2021). The activity space size, rotation and E were calculated based on the standard deviation ellipse (SDE) using jogging trajectory points. Activity space size is the spatial areal size of the ellipse of standard deviation and e represents the ratio of the distances from the center of the ellipse to one of the foci and to one of the vertices of the ellipse (Yuan et al., 2012). The calculation formulas are as follows:

$$A = \sum_{i=1}^n (x_i - \bar{x})^2 - \sum_{i=1}^n (y_i - \bar{y})^2 \quad (6)$$

$$B = \sqrt{A^2 + 4 \left(\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \right)^2} \quad (7)$$

$$C = 2 \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \quad (8)$$

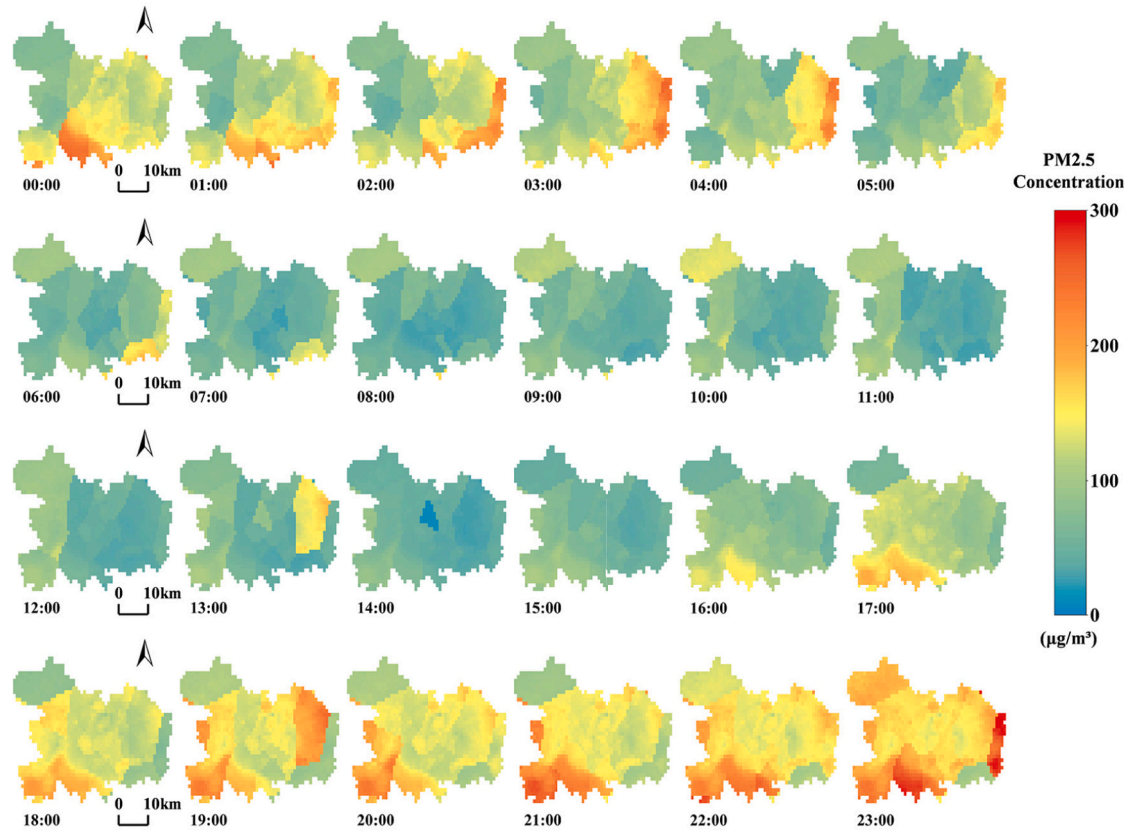
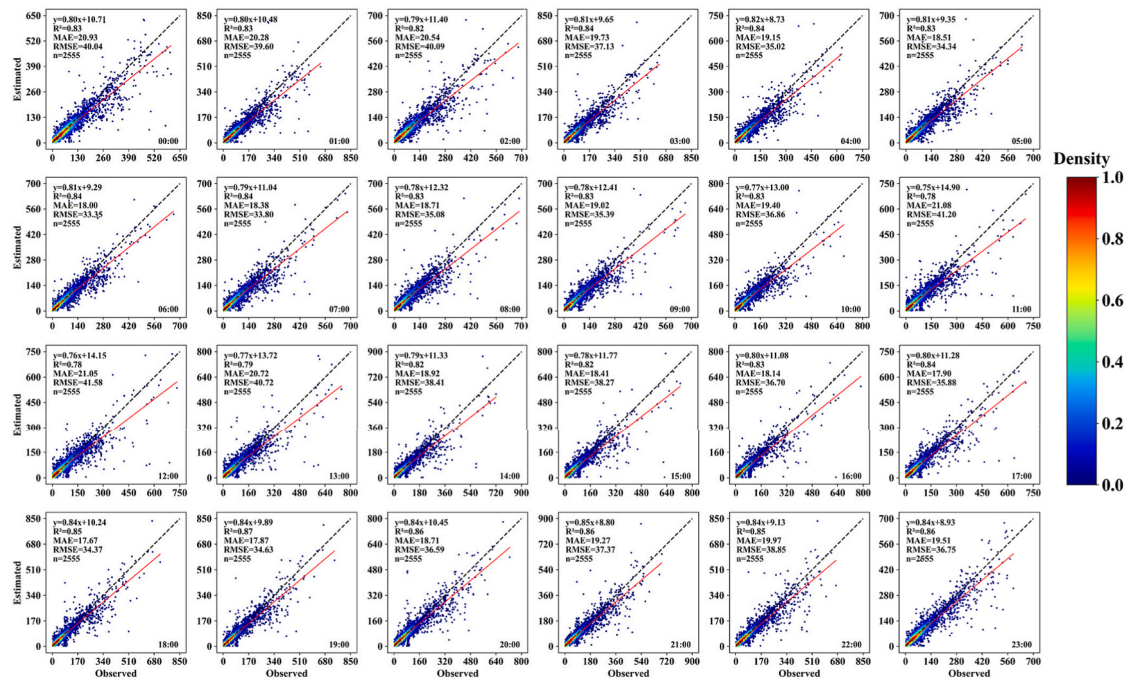
$$Rotation(\theta) = \text{atan} \left(\frac{A+B}{C} \right) \quad (9)$$

$$Size = \pi \cdot a \cdot b \quad (10)$$

$$E = \sqrt{1 - \frac{b^2}{a^2}} \quad (11)$$

where x_i, y_i are the longitude and latitude coordinate of a trajectory point in a trip, respectively; $\bar{x}, \bar{y}$ are the longitude center coordinate and latitude center coordinate of the sampling points $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}, \bar{y} = \frac{\sum_{i=1}^n y_i}{n}$, respectively; θ is the rotation value calculated for a jogging trip; a, b are the $\sqrt{\frac{2 \sum_{i=1}^n ((x_i - \bar{x}) \cos(\theta) - (y_i - \bar{y}) \sin(\theta))^2}{n}}$ and $\sqrt{\frac{2 \sum_{i=1}^n ((x_i - \bar{x}) \sin(\theta) + (y_i - \bar{y}) \cos(\theta))^2}{n}}$, respectively.

Spatial-temporal factors include spatial factors and temporal factors. We selected the six administrative districts within the main urban areas (DC, XC, SJS, FT, HD and CY) in Beijing as location factors in investigating its confounding associations with jogging behavior and PM_{2.5} exposure. The choosing of administrative districts rather than traffic analysis zones and ring road areas is to take advantage of providing potential policy interventions from the government level. Prior works confirmed that outdoor jogging activities of residents have significant

(a) The fused high-resolution $\text{PM}_{2.5}$ data for each hour in a day (Feb 1 2015)(b) The scatter plots of fused hourly $\text{PM}_{2.5}$ concentration and the ground truthFig. 4. The fused $\text{PM}_{2.5}$ concentration data and verification.

temporal differences (Yang et al., 2024b; Yang et al., 2024c). Thus, the temporal factors include the timings of the day, days of the week, and seasons of the year. The timing of jogging is divided into Ante Meridiem 0–12 (am) and Post Meridiem 12–24 (pm). Days of the week for jogging are classified into weekdays and weekends, as there may exist significant differences in jogging patterns between weekdays and weekends (Yang et al., 2023; Huang et al., 2023a; Liu et al., 2022; Tian et al., 2022). It is assumed jogging behavior and related exposure to air pollution differ on a weekday-weekend basis. Four-season circulation was involved to discuss the seasonal effect on jogging exposure to air pollution, as there exists seasonal variation in jogging behavior and potentially its induced exposure to air pollution.

Additionally, gender, the only available personal information via the app, was included as the socio-demographic factor in examining its association with jogging behavior patterns and related exposure to PM_{2.5}. The existing works (Lanza et al., 2020; Zheng et al., 2021; Edwards et al., 2015) confirmed that weather factors, particularly temperature, precipitation, and wind speed, exert a substantial influence on individuals' outdoor physical activities including cycling (Zheng et al., 2021), and walking (Edwards et al., 2015), and may affect their susceptibility to air pollution. Therefore, the city-level weather conditions including these factors are considered exogenous variables in the analysis of jogging behavior patterns and PM_{2.5} exposure. The data were merged with jogging trips based on the time index (Yang et al., 2024b). Table 1 summarizes the statistics of variables included in the model. All the continuous and ordinal variables follow the strict rules of normal distribution (the absolute value of skewness < 3, and that of kurtosis < 10).

3.4. Method

We used aggregative and disaggregated analysis to reveal the jogging behavior patterns and how they relate to joggers' exposure to PM_{2.5}. Aggregative analysis was conducted using descriptive statistics to illustrate the spatial and temporal distribution of jogging behavior patterns and PM_{2.5} exposure. Structural equation modelling (SEM), as the disaggregated analytic method, was later applied to analyze the

complex relationships between PM_{2.5} exposure and jogging behavior patterns. SEM is a statistical method that examines the relationships among numerous variables in a simultaneous way and thus allows prediction of the direct, indirect and total effects of exogenous variables (Huang et al., 2023b; Su and Goulas, 2023). It has merits for this study in illuminating the complex structure of relationships among movement features, spatiotemporal factors, socioeconomic status and PM_{2.5} exposure. Restricted logarithmic likelihood (RLL) was chosen as the estimation method, as most of the observed variables are continuous (following the normal distribution), ordinal or dummy in nature (Kline, 2023).

The path design of the SEM in this study is shown in Fig. 5. We conducted two structural equation models separately for APE₁ and APE₂. The relationships among these variables are delineated as follows. Activity space size is directly shaped by distance and overall trip direction, with the effect mediated by eccentricity and duration (Marquet et al., 2022). Jogging duration correlates with distance, overall direction, and speed, and is further shaped by the eccentricity of the activity space and overall trip direction. Meanwhile, the eccentricity of the activity space is determined by the overall trip direction and speed. These variables in jogging behavior patterns directly contribute to individuals' exposure to PM_{2.5} (Tian et al., 2022). However, the association between jogging behavior patterns and PM_{2.5} exposure is conditioned by spatial and temporal factors (Barreira et al., 2010; Shi et al., 2018). Location is treated as the spatial factor and that poses a direct effect on PM_{2.5} exposure and indirect effects via jogging behavior patterns. Temporal factors from three frames are investigated: timing of the day, weekend/weekend and seasons. These temporal factors have direct effects on jogging behavior pattern and PM_{2.5} exposure, with the effects of temporal factors on PM_{2.5} exposure being mediated by jogging behavior pattern variables. The sole socioeconomic status variable, gender, is considered an exogenous variable impacting jogging behavior pattern variables and PM_{2.5} exposure. Weather conditions, including temperature, precipitation, and wind speed, are designed to pose direct effect on PM_{2.5} exposure and indirect effects via jogging behavior pattern variables. SEM was conducted using the software MPlus 8.3 to investigate

Table 1
Statistics of jogging behavior characteristics and PM_{2.5} exposure.

Variables	Name	Type	Statistics						
			Mean	S.D.	Skewness	Kurtosis	Min	Max	
PM _{2.5} exposure	APE ₁ (µg/m ³)	Numerical	60.43	49.73	1.95	7.95	0.33	745.15	
	APE ₂ (µg)	Numerical	24.28	29.58	3.94	32.02	0.00	861.08	
Behavior pattern characteristics	Speed (km/h)	Numerical	7.66	3.06	−0.03	0.24	3.79	20.09	
	Size (km ²)	Numerical	0.86	1.90	6.81	75.87	0.01	44.56	
	Distance (km)	Numerical	5.22	3.41	0.87	0.59	0.83	18.38	
	Duration (min)	Numerical	44.58	29.39	1.45	3.55	5.00	210.00	
	E	Numerical	0.84	0.16	−1.02	0.37	0.02	1.00	
	Rotation	Categorical	Rotate 90–180° (56.24 %)	0.56	0.50	−0.25	−1.94	–	–
			Rotate 0–90° (43.76 %)(Ref)	0.44				–	–
Temporal factors	Timing	Categorical	AM (35.06 %)	0.35	0.48	0.63	−1.61	–	–
			PM (64.94 %)(Ref)	0.65				–	–
	Day	Categorical	Weekday (74.29 %)	0.74	0.44	−1.11	−0.77	–	–
			Weekend (25.71 %)(Ref)	0.26			−0.77	–	–
	Season	Categorical	Spring (38.02 %)(Ref)	0.38	0.49	0.49	−1.76	–	–
			Summer (35.85 %)	0.36	0.48	0.59	−1.65	–	–
			Autumn (10.25 %)	0.10	0.30	2.62	4.87	–	–
Winter (15.87 %)			0.16	0.37	1.87	1.49	–	–	
Spatial factors	District	Categorical	DC (6.70 %)	0.07	0.25	3.46	9.99	–	–
			XC (7.57 %)	0.08	0.26	3.21	8.30	–	–
			FT (11.68 %)	0.12	0.32	2.39	3.70	–	–
			CY (36.60 %)	0.37	0.48	0.56	−1.70	–	–
			SJS (5.06 %)	0.05	0.22	4.10	14.80	–	–
			HD (32.39 %)(Ref)	0.32	0.47	0.75	−1.43	–	–
			Male (87.87 %)	0.88	0.33	−2.32	3.38	–	–
			Female (12.13 %)(Ref)	0.12				–	–
Socioeconomic status	Gender	Categorical							
Meteorological factors	Temperature (°C)	Numerical	18.21	9.94	−0.72	−0.39	−9.00	38.00	
	Precipitation (mm/h)	Numerical	0.04	0.33	21.76	753.68	0.00	22.46	
	Wind Speed (km/h)	Numerical	10.67	7.14	1.36	2.27	0.00	57.60	

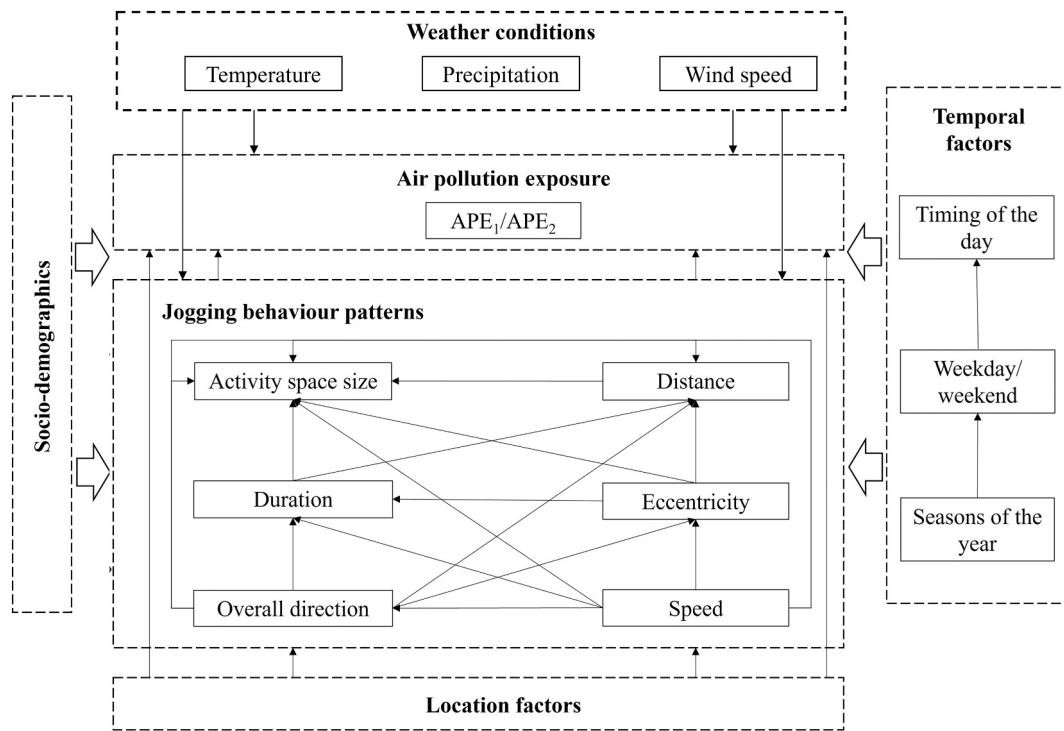


Fig. 5. Path diagram of SEM.

these complex relationships.

4. Results

4.1. Descriptive statistical analysis

4.1.1. Behavior features distribution

The average $PM_{2.5}$ exposure concentration and inhalation doses for jogging are $60.43 \mu g/m^3$ and $24.28 \mu g$ respectively. The box plots in Fig. 6 presents the summary of $PM_{2.5}$ exposure concentration [APE_1] and $PM_{2.5}$ dose inhalation [APE_2] for jogging behavior features across different quartiles. It shows exposure concentration slightly decreases with the increase in activity space size, distance and duration, while dose inhalation increases with the growth in these jogging behavior features. This is probably because of the increased inhalation dose and duration when people took jogging trips in longer distances and larger activity spaces. The reason for the slight decrease in exposure concentration may be due to the complex relationship between spatiotemporal-varied $PM_{2.5}$ distribution and dynamics of jogging behavior. Both exposure concentration and dose inhalation do not seem to vary with the difference in eccentricity and rotation of activity space. Dose inhalation drops with an increase in average jogging speed since joggers tend to travel with longer durations at slower speeds (Wei et al., 2020).

4.1.2. Temporal distribution

We investigated the temporal distribution of $PM_{2.5}$ exposure from three temporal frames: timing of the day, weekday/weekend and across seasons. As shown in Fig. 7, jogging behavior is highly motivated at 5–9 a.m. (morning peak) and 6–10 p.m. (evening peak) of the day and from the end of March to the end of September (Spring and Summer) of the year (Yang et al., 2023; Liu et al., 2023; Tian et al., 2022). More individuals were opting to go for a jog in the post-meridiem period, experiencing significantly reduced exposure to $PM_{2.5}$ compared to those who preferred jogging in the ante-meridiem period, encountering lower levels of $PM_{2.5}$ concentration (by 4.61 %) and inhalation dose (by 13.72 %). This may be partly attributed to the elevated $PM_{2.5}$ concentration in

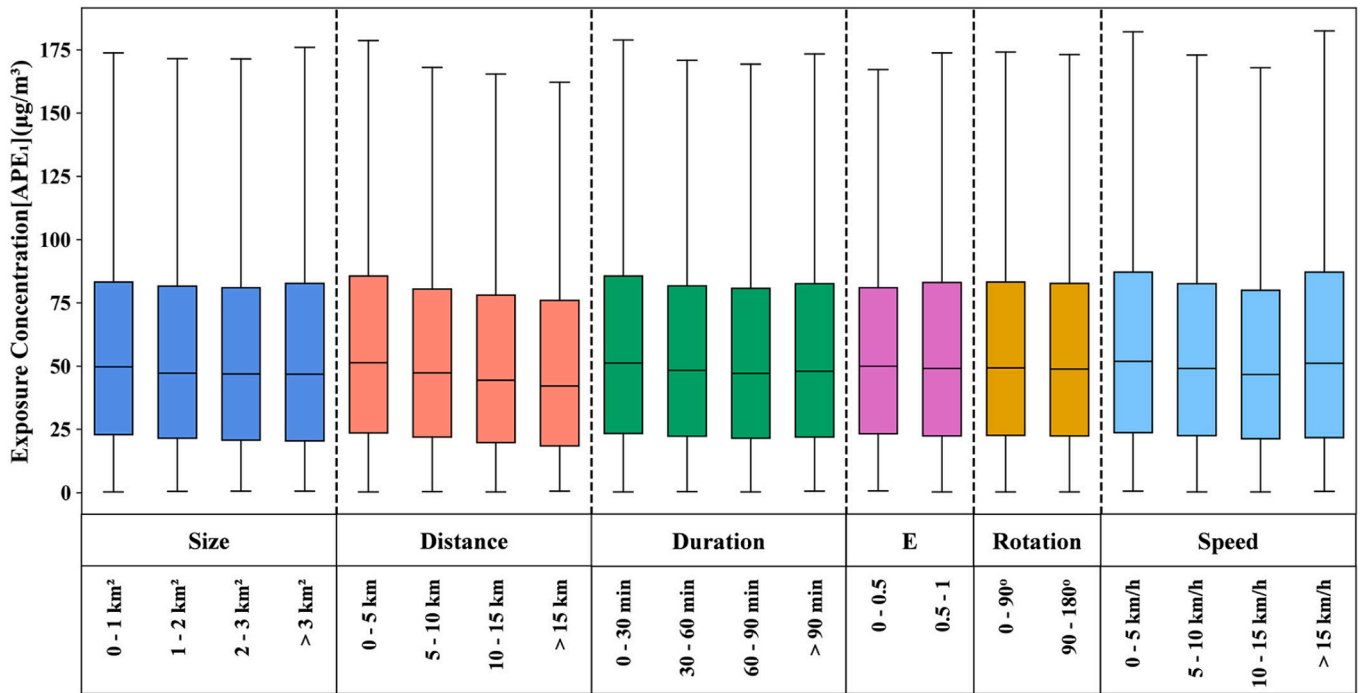
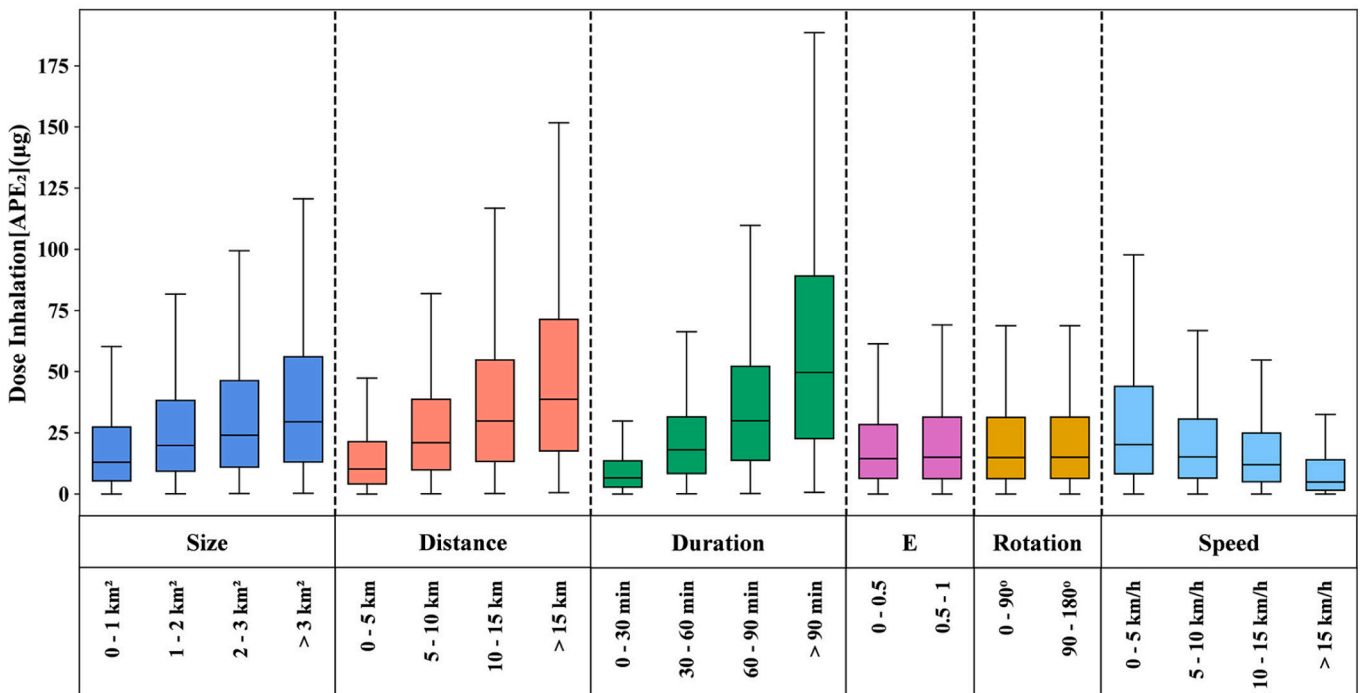
the morning (Table 2). In discussing the weekday/weekend difference, people who took jogging trips on weekdays were exposed to slightly higher levels of $PM_{2.5}$ concentration than those at the weekend, while they inhaled slightly fewer doses of $PM_{2.5}$. This is probably due to the average longer duration of jogging trips during the weekend. Across the seasons of the year 2015, Spring and Summer counted up to 73.87 % of all the jogging trips. Jogging trips in Spring and Summer have significantly lower levels of $PM_{2.5}$ concentration and $PM_{2.5}$ dose inhalation than in Autumn and Winter. People who chose to jog in Winter were exposed to the highest levels of $PM_{2.5}$ risks both in terms of average concentration and dose inhalation. This is perhaps because Spring and Summer are the seasons of better air quality compared with Winter and Autumn (Wei et al., n.d.).

A significant negative relationship exists between jogging frequency and $PM_{2.5}$ concentration across different days of the year, with a coefficient of -0.18 at a significance level of 0.001 . This indicates that there is a higher prevalence of jogging activity on days with lower pollution levels. Furthermore, there is a positive correlation between joggers' inhalation of $PM_{2.5}$ and the duration of their jogging sessions, with a coefficient of 0.21 at a significance level of 0.001 , suggesting a weak correlation between dose inhalation and duration. There is more fluctuation both in terms of $PM_{2.5}$ dose inhalation and jogging duration in Spring and Winter (Fig. 8). This is probably because jogging behavior in these seasons is more constrained by weather conditions (e.g., temperature) (Liu et al., 2023).

4.1.3. Spatial distribution

There exists a significant location difference in jogging behavior patterns and related $PM_{2.5}$ exposure. Jogging trajectories taking place in HD and CY (66.0 % of all trajectories) have the lowest levels of $PM_{2.5}$ exposure concentration and comparatively lower levels of $PM_{2.5}$ dose inhalation in contrast with other districts (Table 3).

Joggers in SJS selected routes characterized by the shortest distance, duration, activity space size, and average speed, leading to their exposure to the lowest levels of dose inhalation. However, jogging trajectories in SJS exhibited the highest levels of $PM_{2.5}$ concentration with

(a) PM_{2.5} exposure concentration [APE₁] for jogging behaviors(b) PM_{2.5} dose inhalation [APE₂] for jogging behaviorsFig. 6. Statistical analysis of PM_{2.5} exposure for jogging behavior features.

those in FT, DC, and XC following suit. This is attributed to the elevated PM_{2.5} levels prevalent within these areas. Given that jogging trajectories in DC and XC are of the longest duration and largest activity space size, they consequently experienced the highest inhalation dose.

4.1.4. Gender difference

There exists a statistically significant difference in PM_{2.5} dose

inhalation (p -value: 0.001), with women inhaling significantly lower doses than men (19.70 μg compared with 24.91 μg). However, there is no significant difference in PM_{2.5} exposure concentration between genders (p -value: 0.176), with women exposed to an average PM_{2.5} concentration of 60.76 $\mu\text{g}/\text{m}^3$, slightly higher than men at 60.38 $\mu\text{g}/\text{m}^3$. This suggests that although women encounter slightly elevated levels of PM_{2.5} concentration, their shorter duration of exposure and lower

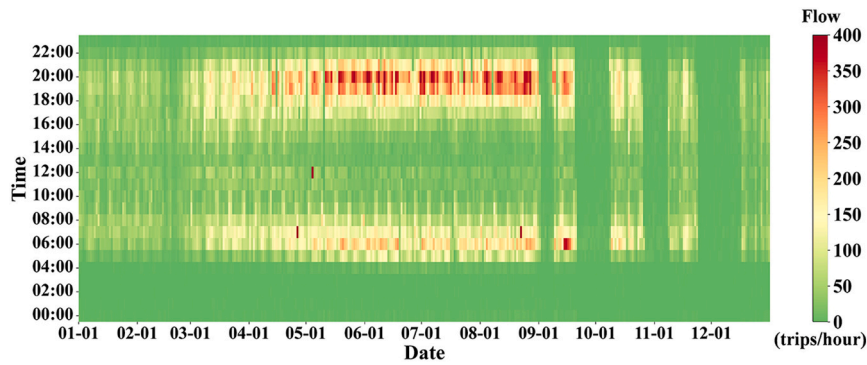


Fig. 7. Temporal distribution of the number of jogging trips (jogging flow).

Table 2
Temporal distribution of PM_{2.5} exposure.

Temporal frames	Categories	APE ₁		APE ₂	
		Mean	S.D.	Mean	S.D.
Timing of the day	A.M.	62.293	51.217	26.653	32.595
	P.M.	59.420	48.872	22.996	27.738
Weekday/Weekend	Weekday	60.712	48.347	24.038	28.214
	Weekend	59.604	53.503	24.972	33.215
Season	Spring	56.151	40.535	21.821	24.226
	Summer	54.816	38.469	22.217	23.434
	Autumn	73.643	58.919	30.677	37.501
	Winter	74.811	74.983	30.686	43.215

inhalation rates contribute to the observed difference. These findings highlight the significance of gender-specific consideration for both individual behaviors and physiological factors in assessing PM_{2.5} exposure during outdoor activities.

4.1.5. Weather conditions

Significant variations in jogging intensity, behavioral patterns, and associated PM_{2.5} exposure are evident across diverse weather conditions (Table 4). Individuals tend to favor jogging on sunny or cloudy (rain-free) days with warm temperatures and gentle breezes. This corresponds with earlier research findings that have identified the connection between weather and outdoor physical activities (Lanza et al., 2020; Zheng et al., 2021). Individuals tend to opt for longer-duration jogs covering greater distances and larger activity spaces in both warm (11–28 °C) and cold (−9–10 °C) weather conditions, as opposed to hot weather (29–38 °C). Additionally, they maintain higher average speeds and traverse more circular activity spaces on days without rain and with light rainfall (precipitation below 2 mm/h), compared to days with moderate (precipitation 2–4 mm/h) or heavy rainfall (precipitation above 4 mm/h). Furthermore, joggers will achieve higher average speeds and cover smaller activity spaces when experiencing light breezes (below 12 km/h) or fresh breezes (12–39 km/h), in comparison with strong breezes (above 39 km/h).

However, joggers typically face increased levels of PM_{2.5} concentration and dose inhalation under rain-free or lightly raining conditions, compared to moderate or heavy rainfall, as well as under calm or light breeze conditions, compared to fresh or strong breezes. These occurrences are likely attributed to the heightened concentration of air pollutants during periods of no rain or light rain and lower wind speed (Lanza et al., 2020). In contrast, joggers are exposed to lower levels of PM_{2.5} concentration and inhale fewer doses during warm weather compared with cold and hot weather conditions.

4.2. Model results

The model results show the path diagram fits the data well (Kline,

2023) with RMSEA at 0.051, CFI at 0.998, and SRMR at 0.007. Table 5 summarizes the direct, indirect and total effects with standardized coefficients among the variables with significance.

4.2.1. The effect of jogging behavior characteristics

In total, there is clear evidence that activity space size, total trip distance, duration, eccentricity (*E*), overall travel direction (rotation), and speed are closely related to joggers' exposure to PM_{2.5} with regard to both APE₁ and APE₂. However, the effects of jogging behavior characteristics on APE₁ and APE₂ are via different casual paths (Fig. 9). The effects of activity space size and total trip distance on PM_{2.5} exposure are mainly attributed to the direct paths. That means the larger activity space size and the shorter total trip distance will directly bring higher levels of risks from PM_{2.5} for both aspects of exposure concentration and dose inhalation as they take jogging activity further away from their residential/working areas where the PM_{2.5} concentration may be higher. People jogging with longer total distances are more likely to move within a flatter activity space with a short duration and thus have lower risks of PM_{2.5} exposure. The lower level of average speed is directly associated with decreased risks of inhalation dose but increased PM_{2.5} exposure concentration. Whereas, lower levels of average speed will significantly increase the duration and further raise the risks of inhalation dose. The indirect effect of speed on inhalation dose via duration with the opposite sign to the direct effect has resulted in a significantly negative total effect.

The longer jogging duration is positively associated with higher levels of PM_{2.5} inhalation doses. In contrast, the effect of duration on APE₁ is mainly attributed to the indirect path via distance, which means the longer duration brings a longer distance and is further associated with lower levels of PM_{2.5} concentration. Although the longer duration of jogging will bring bigger activity space size and indirectly pose slightly higher levels of PM_{2.5} exposure concentration, the coefficient is comparatively smaller than the indirect effect via distance. These result in a negative total effect of duration on APE₁ but a positive total effect on APE₂. People with activity space of a bigger *E* value (flatter activity space) will be exposed to higher levels of average PM_{2.5} concentration. This may be because they are more likely to jog further away from their starting points and be exposed to higher levels of PM_{2.5} concentration. Whereas people with activity space of a smaller *E* value will inhale less air pollution doses, which is attributed to the indirect effect via duration. This means people with flatter activity spaces are more likely to jog for a short duration, rather than jogging around a park or a sports square for a longer duration (Liu et al., 2023; Tian et al., 2022). Overall, the jogging direction poses a total positive effect with significance on APE₂ but not on APE₁. People with jogging behavior patterns following the Southeast-to-Northwest direction inhale higher levels of PM_{2.5} doses. The contradictory direct effect and indirect effects of jogging direction on exposure concentration lead to insignificance of the total effect.

Despite these direct and indirect effects, the association between jogging behavior and joggers' exposure to PM_{2.5} is more complex than

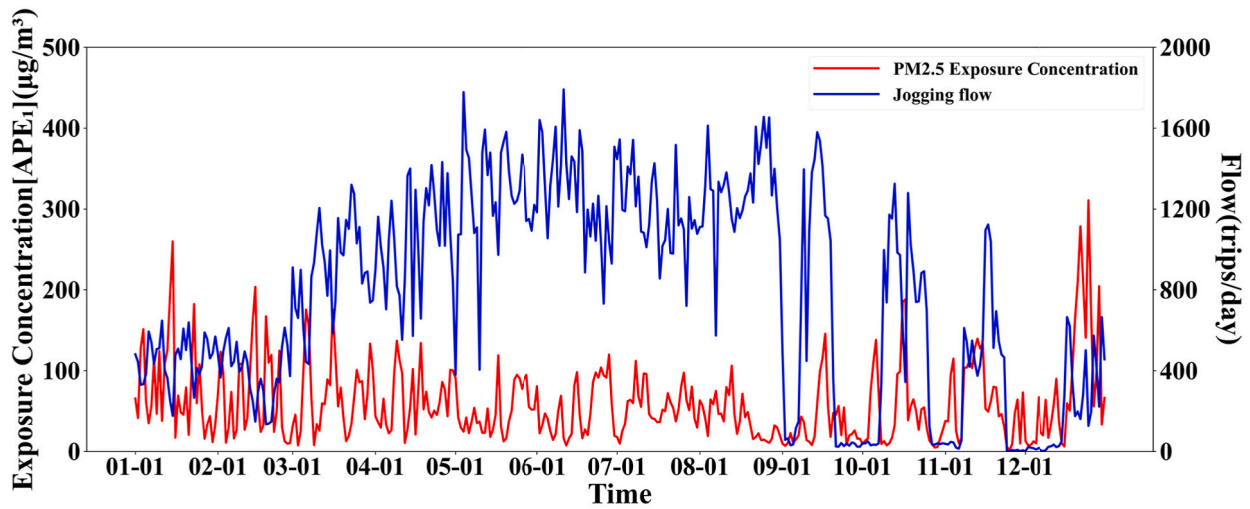
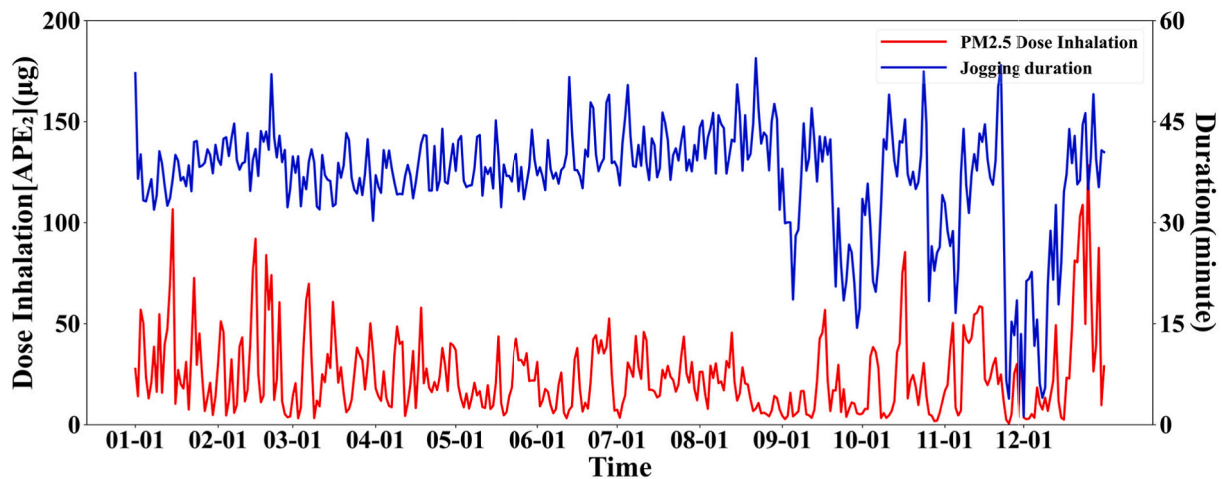
(a) PM_{2.5} exposure concentration and jogging flow(b) PM_{2.5} dose inhalation and jogging durationFig. 8. The relationship between jogging behavior and PM_{2.5} exposure.

Table 3

Location distribution of PM_{2.5} exposure.

Location	Jogging Number	APE ₁	APE ₂	Average Duration	Average Distance	Average Size	Average Speed	Average E
DC	19,283	61.454	26.199	47.094	5.602	1.387	7.708	0.850
XC	21,768	61.414	25.458	46.033	5.105	1.076	7.293	0.863
FT	33,587	62.843	24.108	42.490	4.925	0.709	7.506	0.840
HD	93,178	60.280	23.822	43.575	4.987	0.632	7.684	0.820
CY	105,277	59.033	24.405	46.016	5.676	1.019	7.841	0.841
SJS	14,570	63.050	22.365	39.941	3.855	0.445	7.095	0.843

preconceived as it is conditioned by meteorological factors and is mediated by spatial and temporal factors and may vary by location settings and timing choices.

4.2.2. The effect of temporal factors

The timing choice of jogging demonstrates a more significant effect on average PM_{2.5} exposure density than PM_{2.5} dose inhalation. An individual's timing choice for jogging in the morning is directly associated with a higher risk of PM_{2.5} compared with afternoon and evening. This may be because of the higher levels of PM_{2.5} concentration in the morning due to the large discharge of transport pollutants in the morning peak and the weak solar radiation capacity that is conducive to

the impact of pollutant diffusion (Shi et al., 2018). Whereas, the direct effect on inhalation dose contrasts with the indirect effects primarily mediated by duration, each contributing approximately equally, resulting in insignificance in the total effect.

There are also variations between weekdays and weekend days and across different seasons. These temporal factors also mediate the impact of jogging behavior characteristics on the risks of PM_{2.5} exposure. There is a direct positive effect with the significance of weekday/weekend choice on both APE₁ and APE₂. This is consistent with the fact that people have higher risks of exposure to PM_{2.5} on weekdays (Tong et al., 2020). However, the total effect of weekday/weekend choice on APE₁ and APE₂ could be different when including the effect of jogging

Table 4Jogging behavior and PM_{2.5} exposure under different weather conditions.

Weather conditions	Categories	Jogging number	APE ₁	APE ₂	Average duration	Average distance	Average size	Average speed	Average E
Temperature ^a (°C)	−9–0	16,853	86.574	36.363	46.566	5.497	0.995	7.660	0.863
	0–10	43,552	71.955	29.679	45.609	5.350	0.929	7.598	0.850
	10–28	177,456	55.556	21.951	43.969	5.205	0.843	7.723	0.833
	28–38	49,802	58.857	23.756	45.182	5.093	0.800	7.507	0.830
Precipitation ^b (mm/h)	0	259,728	61.625	24.758	44.444	5.224	0.856	7.678	0.836
	0–2	26,751	49.365	19.929	45.952	5.247	0.870	7.528	0.841
	2–4	821	52.071	16.956	39.798	4.545	0.712	7.454	0.849
	>4	363	37.536	17.711	51.088	5.682	1.001	7.235	0.824
Wind speed ^c (km/h)	0–12	202,054	67.843	27.419	44.712	5.243	0.873	7.655	0.839
	12–39	84,207	43.303	17.015	44.233	5.180	0.819	7.686	0.831
	>39	1402	20.291	7.897	46.268	5.246	0.956	7.371	0.853

Notes:

^a <https://www.cma.gov.cn/>.^b <http://www.chinamsa.org/>.^c <https://std.samr.gov.cn/gb>.

behavior. The total indirect effects of jogging behavior characteristics on APE₁ and APE₂, are negative but with different paths. The indirect effect of jogging behavior characteristics on APE₁ is mainly mediated by timing choices. People tend to choose jogging in the afternoon and evening on weekdays and are exposed to lower levels of PM_{2.5} concentration as there are higher levels of time flexibility after work (Liu et al., 2023; Kan et al., 2023). However, the absolute value of total indirect effects with a negative sign is smaller than the coefficient of the direct effect, resulting in people jogging on weekdays are still exposed to higher levels of PM_{2.5} concentration. In comparison, the absolute value of the total indirect effect of weekday/weekend-day choice on APE₂ approximates that of the direct effect, resulting in the non-significance in the total effect. The indirect effect is made mainly via duration, meaning that people tend to jog for a short duration at the weekend and inhale fewer doses of PM_{2.5}. This significantly reduces the risks of air pollution inhalation when choosing to jog on weekend days.

Seasonal factors will directly affect joggers' exposure to PM_{2.5}. In Autumn and Winter, joggers' exposure risks to PM_{2.5}, both APE₁ and APE₂, are significantly higher than that in Spring, whilst jogging in Summer is exposed to significantly lower levels of PM_{2.5} risks. This is because Autumn and Winter are the most severe seasons of PM_{2.5} and Summer is the season of best air quality in Beijing (Kuerban et al., 2020). Considering the indirect effects via jogging behavior characteristics, there are significant impacts of seasons on APE₂ but limited effects on APE₁. In Summer, the indirect effect on APE₁ through the path via duration and distance with a negative sign slightly enhanced the direct effect of seasonal factors. The longer duration and distance reduced the risks of PM_{2.5} concentration. However, it increased the inhalation of PM_{2.5} significantly and resulted in the insignificance of total effects on APE₂ combined with the direct seasonal effect. In Autumn, people tend to choose jogging more frequently in the morning, which raises the levels of PM_{2.5} concentration, which slightly added to the total positive effects on APE₁, and the longer duration of jogging during Autumn season strengthened the effect on APE₂. Whereas, in Winter, people tend to jog for shorter distances and have lower risks of PM_{2.5} concentration. This slightly reduced the total positive effect of season factors on APE₁. However, the longer duration of jogging is significantly associated with higher levels of PM_{2.5} inhalation, which then reinforces the seasonal effect on APE₂.

4.2.3. The effect of location factors

Six administrative districts were investigated to uncover the location effect of jogging behavior on PM_{2.5} exposure, within which HD was treated as the reference, where the annual air pollution quality is better compared with other districts (Shi et al., 2018). In total, individuals whose jogging behavior with geometric centers located in DC, FT and SJS encounter higher levels of PM_{2.5} concentration, whereas those in CY experience lower levels. Jogging activities occurring in DC, XC, and CY

have higher risks of dose inhalation, while those in SJS have lower risks of dose inhalation. The impact of location factors on PM_{2.5} exposure concentration is primarily ascribed to direct pathways. This means the location factors play a more important role in shaping exposure concentration in these districts despite the indirect effects via jogging behavior characteristics. The jogging behavior characteristics within DC and CY mitigate the negative effects of location on exposure concentration, while those in SJS slightly enhanced the negative effects of location factors.

In comparison, the impact of location factors on dose inhalation in DC, XC and CY, is mainly made through the indirect effects via duration and via activity space size in addition to the direct effects. This means joggers in these districts inhale more PM_{2.5} doses as they choose the longer duration of jogging and larger activity space size, because there are more jogging facilities and places (parks, sports squares, etc.) within these districts (Tian et al., 2022; Wu and Kim, 2021). In FT, the indirect effects via duration and via activity space size are opposite to the direct effect of location, meaning that people in this district have shorter jogging duration and smaller activity space sizes, which may be due to limited jogging infrastructure and location choices (Wu and Kim, 2021). This results in the total effect being negative with significance as the absolute total value of indirect effects is larger than the direct effect. In SJS, the indirect effects via duration and activity space size largely mitigate the direct effect of location on inhalation doses. People within this area with less-developed jogging facilities seem to jog with short duration and smaller activity space size and thus inhale fewer doses of PM_{2.5}.

4.2.4. The effect of gender

Our observations suggest that gender does not correlate with joggers' PM_{2.5} exposure concentration, indicating no significant differences between males and females in this aspect. However, there is a notable association between gender and PM_{2.5} dose inhalation, with female joggers inhaling lower doses compared to their male counterparts. This discrepancy is attributed to both direct and indirect pathways. The direct effect stems from the variance in inhalation rates during physical activities between males and females (Bigazzi and Figliozzi, 2014). Indirect effects primarily operate through pathways involving duration, distance-duration, and duration-timing. Female joggers typically engage in shorter-duration runs and consequently inhale fewer PM_{2.5} doses compared to males. Additionally, female joggers' preference for morning runs contributes to shorter durations, as they often have less flexibility in timing compared to after-work hours (Kan et al., 2023). These shorter durations result in shorter distances covered by female joggers, further associated with reduced levels of PM_{2.5} inhalation doses. This reveals the multifaceted nature of gender differences in PM_{2.5} exposure during jogging activities and highlights the importance of considering various factors, including physiological responses and behavior

Table 5
SEM results.

		APE ₁	APE ₂	Size	Distance	Duration	E	Rotation	Timing
Size	D	0.024***	0.021***						
	T	0.024***	0.021***						
Distance	D	-0.058***	-0.075***	0.427***					
	I	0.010***	0.009***						
Duration	T	-0.047***	-0.066***	0.427***					
	D	0.013***	0.602***	0.058***	0.835***				
E	I	-0.038***	-0.054***	0.356***					
	T	-0.025***	0.548***	0.415***	0.835***				
Rotation	D	0.010***	0.001	0.121***	-0.027***	-0.046***			
	I	0.005***	-0.021***	-0.031***	-0.039***				
Speed	T	0.015***	-0.020***	0.091***	-0.066***	-0.046***			
	D	0.004**	0.003**	0.018***	0.003***	0.023***	-0.081***		
Timing	I	-0.002***	0.015***	0.004***	0.025***	0.004***			
	T	0.002	0.017***	0.022***	0.028***	0.027***	-0.081***		
Weekday	D	-0.020***	0.006**	0.052***	0.611***	-0.340***	-0.246***		
	I	-0.023***	-0.221***	0.097***	-0.268***	0.011***			
Season2	T	-0.043***	-0.215***	0.150***	0.343***	-0.329***	-0.246***		
	D	-0.047***	-0.033***	0.043***	0.013	0.069***	0.081***	-0.003	
Season3	I	0.000	0.036***	0.042***	0.052***	-0.004***	0.000		
	T	-0.048***	0.003	0.085***	0.066***	0.066***	0.082***	-0.003	
Season4	D	0.019***	0.012***	-0.014***	0.005***	-0.023***	0.000	-0.013***	-0.040***
	I	0.002***	-0.014***	-0.011***	-0.022***	-0.003***	-0.002***	0.000	
DC	T	0.021***	-0.002	-0.025***	-0.017***	-0.026***	-0.002	-0.013***	-0.040***
	D	-0.112***	-0.083***	-0.003	0.009***	0.038***	0.018***	-0.009***	0.108***
XC	I	-0.006***	0.020***	0.030***	0.037***	0.006***	0.010***	0.000	
	T	-0.118***	-0.063***	0.027***	0.047***	0.044***	0.028***	-0.009***	0.108***
FT	D	0.064***	0.045***	0.022***	-0.006***	0.025***	0.024***	-0.002	-0.209***
	I	0.010***	0.013***	-0.008***	0.005***	-0.015***	-0.017***	0.001	
CY	T	0.074***	0.058***	0.014***	-0.001	0.010***	0.007***	-0.002	-0.209***
	D	0.134***	0.094***	0.026***	0.003**	0.026***	0.034***	0.009***	-0.420***
SJS	I	0.020***	0.013***	-0.020***	-0.008***	-0.029***	-0.035***	0.001	
	T	0.155***	0.107***	0.006**	-0.005*	-0.003	-0.001	0.010***	-0.420***
Gender	D	0.008***	0.005***	0.072***	0.020***	0.031***	0.044***	-0.012***	0.028***
	I	-0.001***	0.016***	0.027***	0.025***	-0.001***	0.003***	0.000	
Temperature	T	0.007***	0.021***	0.099***	0.044***	0.031***	0.047***	-0.012***	0.028***
	D	0.002	0.000	0.049***	0.013***	0.012***	0.056***	-0.047***	0.029***
Precipitation	I	0.000*	0.005***	0.017***	0.007***	-0.002***	0.006***	0.000	
	T	0.001	0.005***	0.066***	0.020***	0.011***	0.062***	-0.047***	0.029***
Wind Speed	D	0.012***	0.008***	0.013***	0.017***	-0.015***	0.038***	-0.047***	-0.002
	I	0.000**	-0.010***	0.002*	-0.016***	-0.003***	0.004***	0.000	
Intercepts	T	0.012***	-0.003	0.016***	0.001	-0.018***	0.034***	-0.047***	-0.002
	D	-0.004**	-0.006***	0.044***	0.050***	0.048***	0.061***	-0.045***	0.053***
Residual Variances	I	-0.004***	0.022***	0.050***	0.038***	-0.001**	0.008***	0.000	
	T	-0.009***	0.016***	0.094***	0.088***	0.048***	0.069***	-0.045***	0.053***
SJS	D	0.003	-0.001	0.009***	-0.024***	-0.039***	0.014***	-0.051***	0.026***
	I	0.001***	-0.021***	-0.024***	-0.033***	0.000	0.006***	0.000	
Gender	T	0.004**	-0.022***	-0.015***	-0.057***	-0.039***	0.020***	-0.051***	0.026***
	D	0.002	0.053***	-0.002	0.007***	0.067***	0.021***	0.009***	0.047***
Temperature	I	-0.004***	0.036***	0.037***	0.058***	0.002***	0.003***	0.000	
	T	-0.002	0.089***	0.035***	0.064***	0.069***	0.024***	0.009***	0.047***
Precipitation	D	0.028***	0.024***	0.031***	-0.017***	0.004	-0.011***	0.024***	-0.650***
	I	0.032***	0.002	-0.061***	-0.038***	-0.041***	-0.055***	0.002	
Wind Speed	T	0.061***	0.027***	-0.030***	-0.056***	-0.038***	-0.066***	0.026***	-0.650***
	D	-0.025***	-0.021***	0.006***	-0.003***	0.003	0.002	-0.002	-0.036***
Intercepts	I	0.002***	0.002*	-0.003***	0.000	-0.002***	-0.003***	0.000	
	T	-0.023***	-0.019***	0.003	-0.003*	0.000	-0.001	-0.002	-0.036***
Residual Variances	D	-0.340***	-0.239***	0.005***	0.002**	0.025***	0.010***	0.002	-0.151***
	I	0.007***	0.013***	-0.001	0.010***	-0.010***	-0.012***	0.000	
Intercepts	T	-0.333***	-0.227***	0.004**	0.012***	0.014***	-0.002	0.002	-0.151***
		1.604	0.126	-1.204	-1.202	2.268	5.780	1.179	2.210
Residual Variances		0.642	0.640	0.753	0.226	0.884	0.921	0.994	0.790

Note: D, direct effect; I, indirect effect; T, total effect.

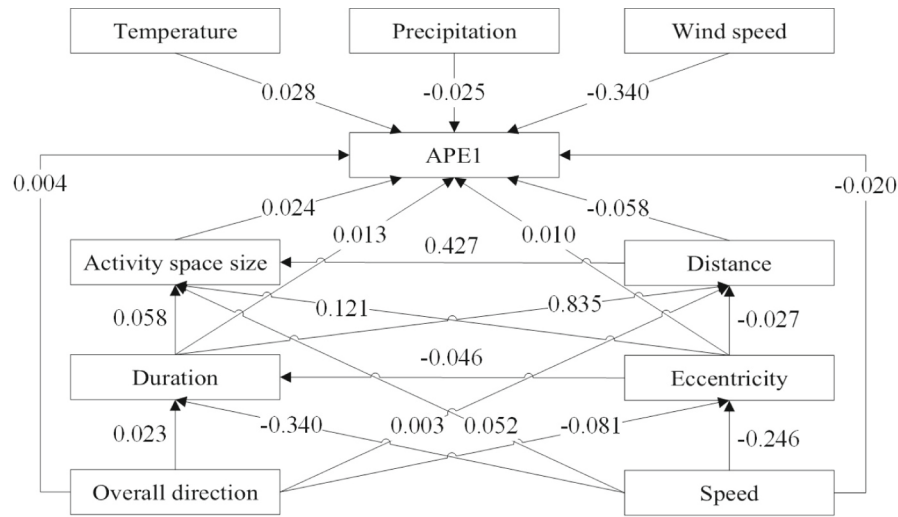
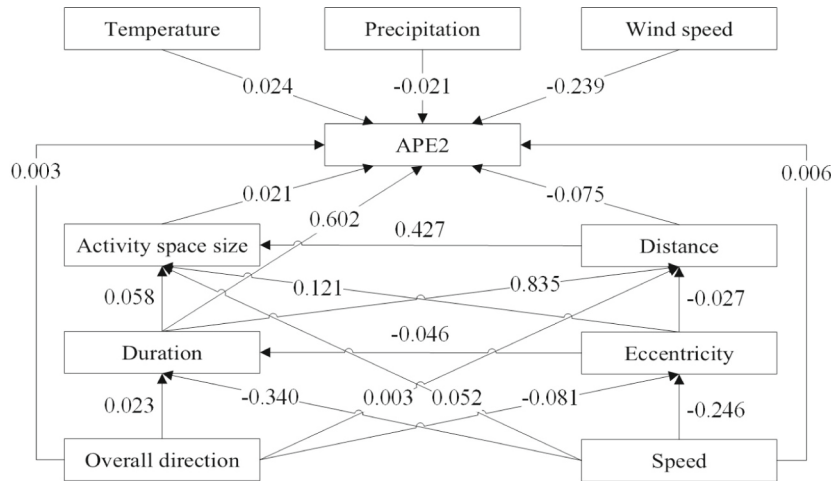
*** Significantly different from zero at $p < 0.01$.** Significantly different from zero at $p < 0.05$.* Significantly different from zero at $p < 0.10$.

patterns, in understanding these disparities. Further research could delve deeper into the underlying mechanisms driving these gender-related differences to inform targeted interventions aimed at minimizing PM_{2.5} exposure risks for all joggers, regardless of gender.

4.2.5. The effect of weather conditions

Weather conditions have statistically significant effects on both APE₁

and APE₂, most of which are mainly attributed to the direct effects, as weather conditions such as temperature, precipitation and wind speed can affect the dispersion, dilution, and accumulation of PM_{2.5} in the atmosphere, and thus directly shapes joggers' PM_{2.5} exposure. Higher temperatures and lower precipitation and lower wind speeds are associated with higher risks of PM_{2.5} exposure. This is in line with previous studies (Lanza et al., 2020; Zheng et al., 2021).

(a) Effects on $PM_{2.5}$ exposure concentration(b) Effects on $PM_{2.5}$ dose inhalation**Fig. 9.** Standardized direct effects of weather conditions and jogging behavior patterns on $PM_{2.5}$ exposure.

However, the effects of weather conditions on joggers' $PM_{2.5}$ exposure are not solely determined by direct effects; it is also influenced by indirect effects through jogging behavior patterns. The indirect effect of temperature on APE₁ primarily operates through timing choices, while its effect on APE₂ is predominantly via timing choices and duration choices. For instance, lower temperatures in the morning are strongly associated with higher concentrations of $PM_{2.5}$. Yet, the shorter duration of jogging sessions in the cooler morning hours results in lower doses of $PM_{2.5}$ inhalation. Consequently, the indirect effects of temperature largely enhance the direct effect on APE₁ while slightly weakening the direct effect on APE₂. In contrast, the indirect effects of precipitation and wind speed slightly attenuate the negative direct effects on both APE₁ and APE₂. When there is low precipitation and wind speed in the morning, it correlates positively with lower levels of $PM_{2.5}$. However, the absolute coefficients for these indirect effects, albeit opposite in sign, are much smaller than those of the direct effects of precipitation and wind speed. This suggests that joggers can potentially mitigate their $PM_{2.5}$ exposure by choosing to jog in the morning when precipitation and wind speeds are low on days characterized by poor air quality.

5. Discussions

5.1. Association between jogging behavior and $PM_{2.5}$ exposure

This study assessed jogging-related $PM_{2.5}$ exposure both in terms of exposure concentration and dose inhalation, and explored the relationships between jogging behaviors and $PM_{2.5}$ exposure in different locations and across three temporal frames. The results show that there exist space-time disparities in joggers' exposure to $PM_{2.5}$. The majority of all jogging trajectories took place in HD and CY and they had the lowest levels of exposure to $PM_{2.5}$ concentration. Jogging trips in the city center (DC and XC) were exposed to the highest levels of both $PM_{2.5}$ concentration and dose inhalation. Joggers had higher risks of $PM_{2.5}$ in Winter and Autumn compared with Summer and Spring, and in the afternoon and evening compared with the morning. Jogging trips on weekdays have a slightly higher risk of exposure to $PM_{2.5}$ concentration but a lower risk of $PM_{2.5}$ dose inhalation than those at the weekend.

It is found clear evidence that joggers' exposure to $PM_{2.5}$ was significantly related to their jogging behavior patterns, including activity space size, distance, overall travel direction, eccentricity and speed. More importantly, there exist systematical relationships between jogging behavior patterns and $PM_{2.5}$ exposure in space and time. The

effects of spatial, temporal and sociodemographic factors as well as weather conditions on $PM_{2.5}$ exposure mediated by jogging behavior characteristics. The direct adverse effects of precipitation and wind speed can be alleviated by the indirect effects mediated through jogging behavior patterns, particularly in terms of timing choices. The effect of location on $PM_{2.5}$ exposure varies across different districts and is mainly mediated by jogging duration and activity space size. The effect of gender on joggers' $PM_{2.5}$ exposure is mediated by duration, distance and timing of the day. In discussing the temporal factors, the effect of weekday/weekend on exposure concentration is mediated by the timing of the day, whilst the effect on dose inhalation is mainly mediated by jogging duration. This proves the effect of weekday/weekend on $PM_{2.5}$ exposure in the medium term is mediated by the timing of the day in the short term. There may exist interconnections across different temporal frames. This could be further explored using different datasets in various geographical contexts, as the findings of this paper are constrained to the specific dataset and location. Moreover, the development of low-cost portable air pollution sensors (e.g., AirBeam, Atmotube, GeoAir) made it possible to directly measure more accurate air pollution exposure than modelling air pollution diffusion based on concentration information from monitoring stations (McKercher et al., 2017). The utilization of air pollution sensors is often combined with travel diary surveys, where participants' sociodemographic information could be acquired in discussing its effect on their air pollution exposure and related socio-environmental disparity issues (Guo et al., 2020; Tao et al., 2021). However, due to its labour-taking nature, the surveys equipped with portable air pollution sensors are hard to apply to a large sample size and over a long period of time.

Voluntary geographic data derived from jogging apps nevertheless offers opportunities to investigate the relationship between jogging behavior patterns and air pollution exposure over a long period. Sociodemographic information more than gender may be also acquired using proper means with the protection of privacy (Cuzzocrea et al., 2022). In addition, the significant negative relationship between jogging flow and $PM_{2.5}$ concentration across the year proves that there may exist self-selection effects (Van Wee and Ettema, 2016), which means health-related attitudes and attitude formation related to air pollution and jogging behavior may condition people's jogging-related air pollution exposure and health outcomes. This is of vital importance in exploring the complex relationship between jogging behavior and air pollution exposure, because attitudes directly shape people's choices of jogging behavior within given geographical environments.

5.2. Policy implications

Our findings have also provided some policy implications. First, implementing effective individual behavior interventions can help to reduce $PM_{2.5}$ exposure (Guo et al., 2020). Generally, choosing to jog in spring and summer, in the afternoon and evening, and at weekends will lower joggers' risk of $PM_{2.5}$ exposure in Beijing. Joggers can decide an appropriate timing according to the air pollution condition on the specific day through broadcast platform of real-time and forecast air pollution information (e.g., Beijing air quality information platform (Beijing Municipal Ecology and Environment Bureau, 2024)). Given the indirect effects of behavior patterns and temporal factors on $PM_{2.5}$ exposure for joggers, policymakers could also consider providing guidance on optimal timing for outdoor physical activities. Encouraging joggers to schedule their runs during times when temperatures and wind speed are moderate, such as early mornings or evenings, may help minimize exposure to elevated pollution levels. Second, developing differentiated strategies for the migration of $PM_{2.5}$ exposure risks for public health interventions. For example, joggers in DC, XC and CY of Beijing could choose smaller activity spaces and shorter durations when jogging, as the indirect paths via activity space and duration will significantly increase their risks of $PM_{2.5}$ exposure to a large proportion. Joggers could choose to do more jogging activities in the afternoon and

evening and on weekend days to avoid the long duration of jogging in highly polluted timing and space. Additionally, males could take more joggings in the morning with shorter distances and durations for the sake of lowering their risks of exposure. Third, from city designers' perspective, urban planning initiatives need to consider air quality (Sun et al., 2017; Song et al., 2019). For instance, measures including designing and implementing sports infrastructure to improve air quality can effectively reduce air pollution exposure of joggers in this area.

6. Conclusion

Physical activity can provide multiple health benefits to individuals in preventing non-communicable diseases and improving quality of life as well as generating additional returns on investment to the civil society, including reduced use of fossil fuels, cleaner air and less congested, safer roads. Air pollution poses nonnegligible threats to people's health during jogging activities. It is also the government's duty to establish physical activity-friendly environments (e.g., friendly facilities, better air quality) for joggers with consideration of their behavior patterns. Joggers' exposure to $PM_{2.5}$ is a combination result of both dynamics of air pollution diffusion and jogging behavior patterns across time and space. Taking Beijing as a case study, we assessed joggers' exposure to $PM_{2.5}$ in terms of both exposure concentration and inhalation dose and uncovered the systematic relationship between jogging behavior patterns and $PM_{2.5}$ exposure in different locations and across different temporal frames. We highlighted that: (1) female joggers inhaled significantly less $PM_{2.5}$ dose than men. (2) There existed significant disparities in joggers' exposure to $PM_{2.5}$ across locations and timings of the day, days of the week and seasons. (3) The effects of spatial, temporal and gender factors as well as weather conditions on $PM_{2.5}$ exposure were mediated by jogging behavior characteristics, especially activity space size, distance, rotation and duration. These findings have merits in providing personal guidance and behavior intervention for individuals regarding jogging choices of routes, distances, locations and timings. This study also supports characterizing jogging behavior patterns in discussing their relationship with behavior-induced environmental health problems.

Despite insightful findings, this study also has certain limitations due to data constraints. (1) Unlike questionnaire surveys, trajectory data from sports apps lacks (or is incomplete) individual demographic attributes, e.g., age, income, and education, since data unavailability and privacy protection (Yang et al., 2023; Liu et al., 2023; Yang et al., 2024c). Although this paper used a random forest model to complete gender information and achieved good results, the completed data also faces issues of inaccuracy. Moreover, crowdsourced datasets are still not "immune" to big data's common limitations including the representativeness of crowdsourcing contributors and selection biases (Yang et al., 2024c; Zheng et al., 2024), etc. Jogging trajectories are consistently skewed because not all jogger groups are equally inclined to share details about their outdoor jogging on sports Apps. Integrating "small data", such as survey data, into big data analysis may offer a promising way to overcome these limitations in future research (Liu et al., 2023). (2) The user profile analysis of the Chinese sports and fitness App market indicates that users aged 18–45 comprise up to 95.6 % (Chinese Academy of Industrial Economy, 2018). Although the Chinese Exposure Factors Handbook specifies that the variance in inhalation rates between individuals aged 18–45 and those aged 45–60 is approximately 1.03 %, with a corresponding 2.53 % shift in the male-to-female ratio (Duan, 2015), multiplying respiratory rates specific to gender without considering factors such as age, physical condition, and environmental context may introduce inaccuracies in the estimation process. A more nuanced approach, taking into account individual sociodemographic and physiological status together with jogging behaviors, is necessary to accurately assess and mitigate the health risks associated with air pollution exposure during outdoor physical activities. (3) Although the fused $PM_{2.5}$ data meets the research requirements ($R^2 = 0.82$), some issues like

abrupt changes in PM_{2.5} concentrations in specific areas (Fig. 4(a)) should be noticed. Future research should invite other updated modeling methods or portable sensors to provide even more precise exposure measurements. (4) The findings are limited to the case study, and they may not apply to other cities and geo-environmental contexts. Future studies on the association between jogging behavior and air pollution need to examine the relationship in different contexts.

CRedit authorship contribution statement

Wenbo Guo: Writing – original draft, Supervision, Resources, Conceptualization. **Jiawei He:** Writing – original draft, Software, Methodology, Investigation, Data curation. **Wei Yang:** Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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